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JEE MAIN · QUESTION PAPER

JEE Main 2022 (Online) 28th June Evening Shift

JEE Main Question Paper with Answer Key

28 Jun 2022	90	13	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

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Mathematics	28 questions
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Out of syllabus	13 questions
Answer key	at the end

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Chemistry 21 questions

Q1. MCQ (Single Correct Answer)

Compound A contains 8.7% Hydrogen, 74% Carbon and 17.3% Nitrogen. The molecular formula of the compound is,

Given : Atomic masses of C, H and N are 12, 1 and 14 amu respectively.

The molar mass of the compound A is 162 g mol^{-1}.

Open solution

(A) $C_4H_6N_2$ **(B)** C_2H_3N **(C)** C_5H_7N **(D)** $C_{10}H_{14}N_2$

Q2. MCQ (Single Correct Answer)

Consider the following statements :

(A) The principal quantum number 'n' is a positive integer with values of 'n' = 1, 2, 3, ...

(B) The azimuthal quantum number 'l' for a given 'n' (principal quantum number) can have values as 'l' = 0, 1, 2, ... n

(C) Magnetic orbital quantum number 'm_l' for a particular 'l' (azimuthal quantum number) has (2l + 1) values.

(D) $\pm$ 1/2 are the two possible orientations of electron spin.

(E) For l = 5, there will be a total of 9 orbital

Which of the above statements are correct?

Open solution

(A) (A), (B) and (C) **(B)** (A), (C), (D) and (E)

(C) (A), (C) and (D) **(D)** (A), (B), (C) and (D)

Q3. MCQ (Single Correct Answer)

In the structure of SF_4, the lone pair of electrons on S is in.

Open solution

(A) equatorial position and there are two lone pair - bond pair repulsions at 90°.

(B) equatorial position and there are three lone pair - bond pair repulsions at 90°.

(C) axial position and there are three lone pair - bond pair repulsion at 90°.

(D) axial position and there are two lone pair - bond pair repulsion at 90°.

Q4. MCQ (Single Correct Answer)

A student needs to prepare a buffer solution of propanoic acid and its sodium salt with pH 4. The ratio of $\frac{[CH_3CH_2COO^-]}{[CH_3CH_2COOH]}$ required to make buffer is _____.

Given : $K_a(CH_3CH_2COOH) = 1.3 \times 10^{-5}$

Open solution

(A) 0.03 **(B)** 0.13 **(C)** 0.23 **(D)** 0.33

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Q5. MCQ (Single Correct Answer)

Open solution

Match List - I with List - II :

	List-I (Oxide)		List-II (Nature)
(A)	Cl_2O_7	(I)	Amphoteric
(B)	Na_2O	(II)	Basic
(C)	Al_2O_3	(III)	Neutral
(D)	N_2O	(IV)	Acidic

Choose the correct answer from the options given below :

(A) (A) - (IV), (B) - (III), (C) - (I), (D) - (II)

(B) (A) - (IV), (B) - (II), (C) - (I), (D) - (III)

(C) (A) - (II), (B) - (IV), (C) - (III), (D) - (I)

(D) (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

Q6. MCQ (Single Correct Answer)

Open solution

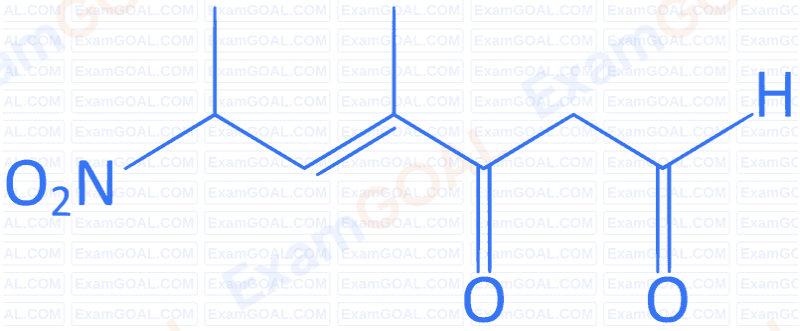
Among the following, basic oxide is :

(A) SO_3 **(B)** SiO_2 **(C)** CaO **(D)** Al_2O_3

Q7. MCQ (Single Correct Answer)

Open solution

The correct IUPAC name of the following compound is :



(A) 4-methyl-2-nitro-5-oxohept-3-enal

(B) 4-methyl-5-oxo-2-nitrohept-3-enal

(C) 4-methyl-6-nitro-3-oxohept-4-enal

(D) 6-formyl-4-methyl-2-nitrohex-3-enal

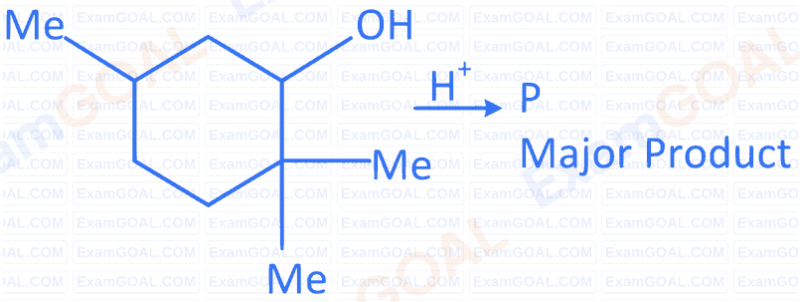
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Q8. MCQ (Single Correct Answer)

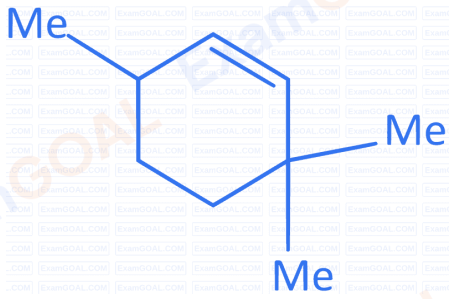
Open solution

The major product (P) of the given reaction is

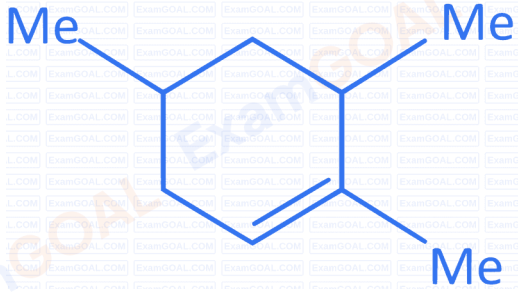
(where, Me is $-CH_3$)



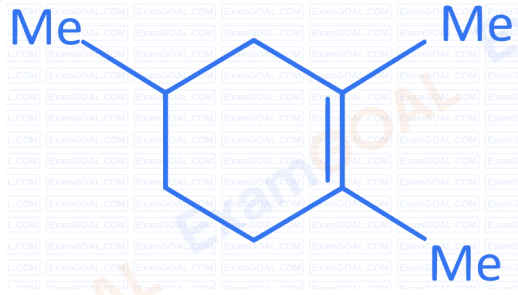
(A)



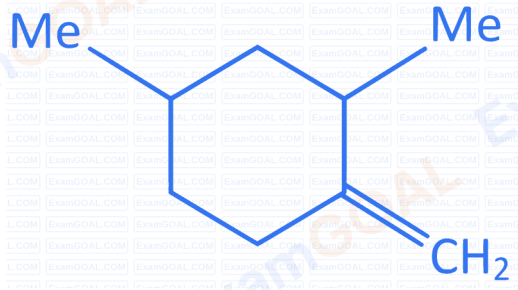
(B)



(C)



(D)



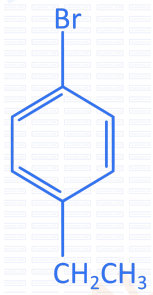
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Q9. MCQ (Single Correct Answer)

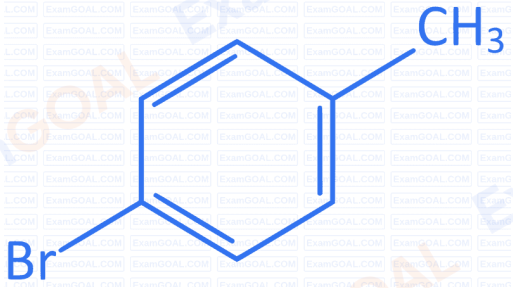
Open solution

$$A \xrightarrow[(ii)\ CN^-]{(i)\ Cl_2, \Delta} \xrightarrow[(iii)\ H_2O / H^+]{} \text{4-Bromophenyl acetic acid.}$$

In the above reaction 'A' is

(A) 

(B) 

(C) 

(D) 

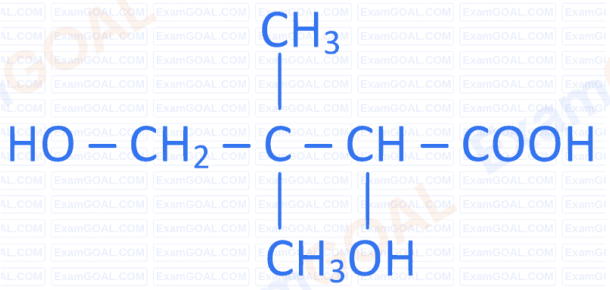
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Q10. MCQ (Single Correct Answer)

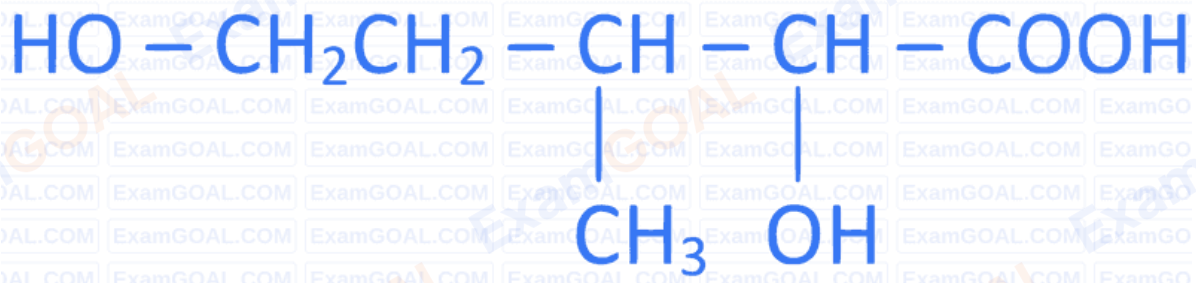
Open solution

Isobutyraldehyde on reaction with formaldehyde and K_2CO_3 gives compound 'A'. Compound 'A' reacts with KCN and yields compound 'B', which on hydrolysis gives a stable compound 'C'. The compound 'C' is

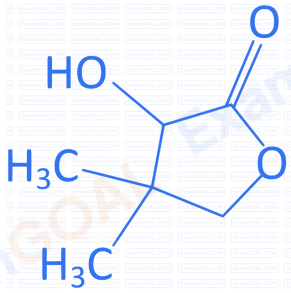
(A)



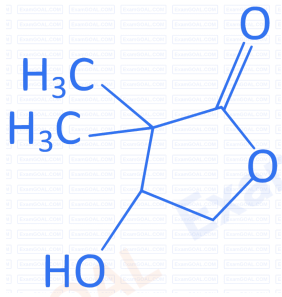
(B)



(C)



(D)

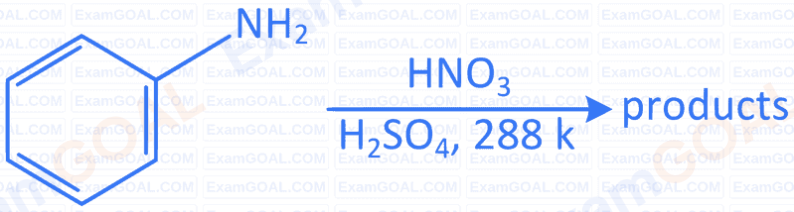


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Q11. MCQ (Single Correct Answer)

Open solution

With respect to the following reaction, consider the given statements :



(A) o-Nitroaniline and p-nitroaniline are the predominant products.

(B) p-Nitroaniline and m-nitroaniline are the predominant products.

(C) HNO_3 acts as an acid.

(D) H_2SO_4 acts as an acid.

Choose the correct option.

(A) (A) and (C) are correct statements.

(B) (A) and (D) are correct statements.

(C) (B) and (D) are correct statements.

(D) (B) and (C) are correct statements.

Q12. MCQ (Single Correct Answer)

Open solution

When sugar 'X' is boiled with dilute H_2SO_4 in alcoholic solution, two isomers 'A' and 'B' are formed. 'A' on oxidation with HNO_3 yields saccharic acid where as 'B' is laevorotatory. The compound 'X' is :

(A) Maltose **(B)** Sucrose **(C)** Lactose **(D)** Starch

Q13. Numerical

Open solution

For combustion of one mole of magnesium in an open container at 300 K and 1 bar pressure, $\Delta_C H^\Theta = -601.70$ kJ mol^{-1}, the magnitude of change in internal energy for the reaction is _____ kJ. (Nearest integer)

(Given : R = 8.3 J K^{-1} mol^{-1})

Q14. Numerical

Open solution

2.5 g of protein containing only glycine ($C_2H_5NO_2$) is dissolved in water to make 500 mL of solution. The osmotic pressure of this solution at 300 K is found to be 5.03×10^{-3} bar. The total number of glycine units present in the protein is _____.

(Given : R = 0.083 L bar K^{-1} mol^{-1})

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Q15. Numerical

For the given reactions

$Sn^{2+} + 2e^- \rightarrow Sn$

$Sn^{4+} + 4e^- \rightarrow Sn$

the electrode potentials are ; $E^o_{Sn^{2+}/Sn} = -0.140$ V and $E^o_{Sn^{4+}/Sn} = +0.010$ V. The magnitude of standard electrode potential for Sn^{4+}/Sn^{2+} i.e. $E^o_{Sn^{4+}/Sn^{2+}}$ is _____ $\times 10^{-2}$ V. (Nearest integer)

Open solution

Q16. Numerical

A radioactive element has a half life of 200 days. The percentage of original activity remaining after 83 days is _____. (Nearest integer)

(Given : antilog 0.125 = 1.333, antilog 0.693 = 4.93)

Open solution

Q17. Numerical

$[Fe(CN)_6]^{4-}$

$[Fe(CN)_6]^{3-}$

$[Ti(CN)_6]^{3-}$

$[Ni(CN)_4]^{2-}$

$[Co(CN)_6]^{3-}$

Among the given complexes, number of paramagnetic complexes is _____.

Open solution

Q18. Numerical

(a) $CoCl_3.4NH_3$, (b) $CoCl_3.5NH_3$, (c) $CoCl_3.6NH_3$ and (d) $CoCl(NO_3)_2.5NH_3$.

Number of complex(es) which will exist in cis-trans form is/are _____.

Open solution

Q19. Numerical

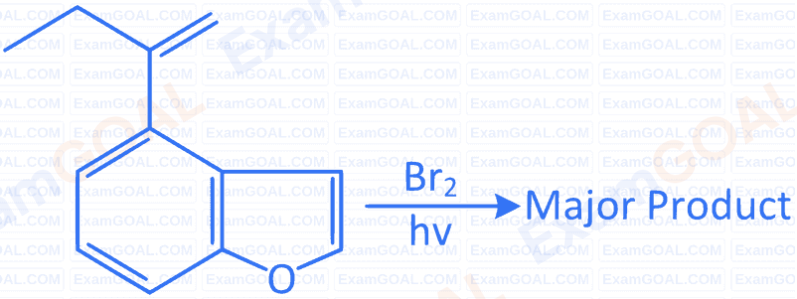
The complete combustion of 0.492 g of an organic compound containing 'C', 'H' and 'O' gives 0.793 g of CO_2 and 0.442 g of H_2O. The percentage of oxygen composition in the organic compound is _____. (nearest integer)

Open solution

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Q20. Numerical

The major product of the following reaction contains _____ bromine atom(s).



Open solution

Q21. Numerical

0.01 M $KMnO_4$ solution was added to 20.0 mL of 0.05 M Mohr's salt solution through a burette. The initial reading of 50 mL burette is zero. The volume of $KMnO_4$ solution left in the burette after the end point is _____ mL. (nearest integer)

Open solution

Mathematics 28 questions

Q22. MCQ (Single Correct Answer)

Let $R_1 = \{(a, b) \in N \times N : |a - b| \leq 13\}$ and

$R_2 = \{(a, b) \in N \times N : |a - b| \neq 13\}$. Then on N :

(A) Both R_1 and R_2 are equivalence relations

(B) Neither R_1 nor R_2 is an equivalence relation

(C) R_1 is an equivalence relation but R_2 is not

(D) R_2 is an equivalence relation but R_1 is not

Open solution

Q23. MCQ (Single Correct Answer)

Let f(x) be a quadratic polynomial such that $f(-2) + f(3) = 0$. If one of the roots of $f(x) = 0$ is -1, then the sum of the roots of $f(x) = 0$ is equal to :

(A) $\frac{11}{3}$ (B) $\frac{7}{3}$ (C) $\frac{13}{3}$ (D) $\frac{14}{3}$

Open solution

Q24. MCQ (Single Correct Answer)

The number of ways to distribute 30 identical candies among four children C_1, C_2, C_3 and C_4 so that C_2 receives at least 4 and at most 7 candies, C_3 receives at least 2 and at most 6 candies, is equal to :

(A) 205 (B) 615 (C) 510 (D) 430

Open solution

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Q25. MCQ (Single Correct Answer)

The term independent of x in the expansion of

$(1-x^2+3x^3)\left(\frac{5}{2}x^3-\frac{1}{5x^2}\right)^{11}$, $x \neq 0$ is :

(A) $\frac{7}{40}$ (B) $\frac{33}{200}$ (C) $\frac{39}{200}$ (D) $\frac{11}{50}$

Open solution

Q26. MCQ (Single Correct Answer)

If n arithmetic means are inserted between a and 100 such that the ratio of the first mean to the last mean is 1 : 7 and a + n = 33, then the value of n is :

(A) 21 (B) 22 (C) 23 (D) 24

Open solution

Q27. MCQ (Single Correct Answer)

Let f, g : R $\rightarrow$ R be functions defined by

$f(x)=\begin{cases}[x] & , \quad x<0 \\ |1-x| & , \quad x \geq 0\end{cases}$ and $g(x)=\begin{cases}e^x-x & , \quad x<0 \\ (x-1)^2-1 & , \quad x \geq 0\end{cases}$ where [x] denote the greatest integer less than or equal to x. Then, the function fog is discontinuous at exactly :

(A) one point (B) two points (C) three points (D) four points

Open solution

Q28. MCQ (Single Correct Answer)

Let f : R $\rightarrow$ R be a differentiable function such that $f\left(\frac{\pi}{4}\right)=\sqrt{2}$, $f\left(\frac{\pi}{2}\right)=0$ and $f'\left(\frac{\pi}{2}\right)=1$ and

let $g(x)=\int_x^{\pi/4}\left(f'(t)\sec t+\tan t \sec t\, f(t)\right)dt$ for $x \in \left[\frac{\pi}{4}, \frac{\pi}{2}\right)$. Then $\lim\limits_{x \to \left(\frac{\pi}{2}\right)^-} g(x)$ is equal to :

(A) 2 (B) 3 (C) 4 (D) -3

Open solution

Q29. MCQ (Single Correct Answer)

Let f : R $\rightarrow$ R be a continuous function satisfying f(x) + f(x + k) = n, for all x $\in$ R where k > 0 and n is a positive integer. If $I_1=\int\limits_0^{4nk} f(x)dx$ and $I_2=\int\limits_{-k}^{3k} f(x)dx$, then :

(A) $I_1+2I_2=4nk$ (B) $I_1+2I_2=2nk$ (C) $I_1+nI_2=4n^2k$ (D) $I_1+nI_2=6n^2k$

Open solution

Q30. MCQ (Single Correct Answer)

The area of the bounded region enclosed by the curve

$y=3-\left|x-\frac{1}{2}\right|-|x+1|$ and the x-axis is :

(A) $\frac{9}{4}$ (B) $\frac{45}{16}$ (C) $\frac{27}{8}$ (D) $\frac{63}{16}$

Open solution

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Q31. MCQ (Single Correct Answer)

Let x = x(y) be the solution of the differential equation

$2y\,e^{x/y^2}dx + \left(y^2 - 4xe^{x/y^2}\right)dy = 0$ such that x(1) = 0. Then, x(e) is equal to :

Open solution

(A) $e\log_e(2)$ (B) $-e\log_e(2)$ (C) $e^2\log_e(2)$ (D) $-e^2\log_e(2)$

Q32. MCQ (Single Correct Answer)

Let the slope of the tangent to a curve y = f(x) at (x, y) be given by $2\tan x(\cos x - y)$. If the curve passes through the point $\left(\frac{\pi}{4}, 0\right)$, then the value of $\int_0^{\pi/2} y\,dx$ is equal to :

Open solution

(A) $(2-\sqrt{2}) + \frac{\pi}{\sqrt{2}}$ (B) $2 - \frac{\pi}{\sqrt{2}}$

(C) $(2+\sqrt{2}) + \frac{\pi}{\sqrt{2}}$ (D) $2 + \frac{\pi}{\sqrt{2}}$

Q33. MCQ (Single Correct Answer)

Let a triangle be bounded by the lines L_1 : 2x + 5y = 10; L_2 : $-$4x + 3y = 12 and the line L_3, which passes through the point P(2, 3), intersects L_2 at A and L_1 at B. If the point P divides the line-segment AB, internally in the ratio 1 : 3, then the area of the triangle is equal to :

Open solution

(A) $\frac{110}{13}$ (B) $\frac{132}{13}$ (C) $\frac{142}{13}$ (D) $\frac{151}{13}$

Q34. MCQ (Single Correct Answer)

Let a > 0, b > 0. Let e and l respectively be the eccentricity and length of the latus rectum of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. Let e' and l' respectively be the eccentricity and length of the latus rectum of its conjugate hyperbola. If $e^2 = \frac{11}{14}l$ and $(e')^2 = \frac{11}{8}l'$, then the value of $77a + 44b$ is equal to :

Open solution

(A) 100 (B) 110 (C) 120 (D) 130

Q35. MCQ (Single Correct Answer)

Let $\vec{a} = \alpha\hat{i} + 2\hat{j} - \hat{k}$ and $\vec{b} = -2\hat{i} + \alpha\hat{j} + \hat{k}$, where $\alpha \in R$. If the area of the parallelogram whose adjacent sides are represented by the vectors $\vec{a}$ and $\vec{b}$ is $\sqrt{15(\alpha^2+4)}$, then the value of $2\left|\vec{a}\right|^2 + \left(\vec{a}.\vec{b}\right)\left|\vec{b}\right|^2$ is equal to :

Open solution

(A) 10 (B) 7 (C) 9 (D) 14

Q36. MCQ (Single Correct Answer)

If vertex of a parabola is (2, $-$1) and the equation of its directrix is 4x $-$ 3y = 21, then the length of its latus rectum is :

Open solution

(A) 2 (B) 8 (C) 12 (D) 16

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Q37. MCQ (Single Correct Answer)

The probability that a randomly chosen one-one function from the set {a, b, c, d} to the set {1, 2, 3, 4, 5} satisfies f(a) + 2f(b) − f(c) = f(d) is :

(A) $\frac{1}{24}$ (B) $\frac{1}{40}$ (C) $\frac{1}{30}$ (D) $\frac{1}{20}$

Open solution

Q38. MCQ (Single Correct Answer)

The value of

$\lim\limits_{n\to\infty} 6\tan\left\{\sum\limits_{r=1}^{n}\tan^{-1}\left(\frac{1}{r^2+3r+3}\right)\right\}$ is equal to :

(A) 1 (B) 2 (C) 3 (D) 6

Open solution

Q39. MCQ (Single Correct Answer)

Let $\vec{a}$ be a vector which is perpendicular to the vector $3\hat{i}+\frac{1}{2}\hat{j}+2\hat{k}$. If $\vec{a}\times\left(2\hat{i}+\hat{k}\right)=2\hat{i}-13\hat{j}-4\hat{k}$, then the projection of the vector $\vec{a}$ on the vector $2\hat{i}+2\hat{j}+\hat{k}$ is :

(A) $\frac{1}{3}$ (B) 1 (C) $\frac{5}{3}$ (D) $\frac{7}{3}$

Open solution

Q40. MCQ (Single Correct Answer)

If cotα = 1 and sec$\beta = -\frac{5}{3}$, where $\pi<\alpha<\frac{3\pi}{2}$ and $\frac{\pi}{2}<\beta<\pi$, then the value of $\tan(\alpha+\beta)$ and the quadrant in which $\alpha + \beta$ lies, respectively are :

(A) $-\frac{1}{7}$ and IVth quadrant (B) 7 and Ist quadrant

(C) -7 and IVth quadrant (D) $\frac{1}{7}$ and Ist quadrant

Open solution

Q41. Numerical

Let the image of the point P(1, 2, 3) in the line $L:\frac{x-6}{3}=\frac{y-1}{2}=\frac{z-2}{3}$ be Q. Let R (α, β, γ) be a point that divides internally the line segment PQ in the ratio 1 : 3. Then the value of 22 $(\alpha + \beta + \gamma)$ is equal to _____.

Open solution

Q42. Numerical

Suppose a class has 7 students. The average marks of these students in the mathematics examination is 62, and their variance is 20. A student fails in the examination if he/she gets less than 50 marks, then in worst case, the number of students can fail is _____.

Open solution

Q43. Numerical

If one of the diameters of the circle $x^2+y^2-2\sqrt{2}x-6\sqrt{2}y+14=0$ is a chord of the circle $(x-2\sqrt{2})^2+(y-2\sqrt{2})^2=r^2$, then the value of r^2 is equal to _____.

Open solution

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Q44. Numerical

If $\lim\limits_{x\to 1}\frac{\sin(3x^2-4x+1)-x^2+1}{2x^3-7x^2+ax+b}=-2$, then the value of (a − b) is equal to _____.

Open solution

Q45. Numerical

Let for n = 1, 2, ..., 50, S_n be the sum of the infinite geometric progression whose first term is n^2 and whose common ratio is $\frac{1}{(n+1)^2}$. Then the value of

$\frac{1}{26}+\sum\limits_{n=1}^{50}\left(S_n+\frac{2}{n+1}-n-1\right)$ is equal to _____.

Open solution

Q46. Numerical

If the system of linear equations
$2x-3y=\gamma+5$,
$\alpha x+5y=\beta+1$, where $\alpha, \beta, \gamma \in$ R has infinitely many solutions then the value of | 9α + 3β + 5γ | is equal to _____.

Open solution

Q47. Numerical

Let $A=\begin{pmatrix}1+i & 1\\ -i & 0\end{pmatrix}$ where $i=\sqrt{-1}$. Then, the number of elements in the set { n ∈ {1, 2, ..., 100} : A^n = A } is _____.

Open solution

Q48. Numerical

Sum of squares of modulus of all the complex numbers z satisfying $\overline{z}=iz^2+z^2-z$ is equal to _____.

Open solution

Q49. Numerical

Let S = {1, 2, 3, 4}. Then the number of elements in the set { f : S × S → S : f is onto and f (a, b) = f (b, a) ≥ a ∀ (a, b) ∈ S × S } is _____.

Open solution

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Physics 28 questions

Q50. MCQ (Single Correct Answer)

Velocity (v) and acceleration (a) in two systems of units 1 and 2 are related as $v_2 = \frac{n}{m^2}v_1$ and $a_2 = \frac{a_1}{mn}$ respectively. Here m and n are constants. The relations for distance and time in two systems respectively are :

Open solution

(A) $\frac{n^3}{m^3}L_1 = L_2$ and $\frac{n^2}{m}T_1 = T_2$

(B) $L_1 = \frac{n^4}{m^2}L_2$ and $T_1 = \frac{n^2}{m}T_2$

(C) $L_1 = \frac{n^2}{m}L_2$ and $T_1 = \frac{n^4}{m^2}T_2$

(D) $\frac{n^2}{m}L_1 = L_2$ and $\frac{n^4}{m^2}T_1 = T_2$

Q51. MCQ (Single Correct Answer)

A ball is spun with angular acceleration α = $6t^2 - 2t$ where t is in second and α is in rads^{-2}. At t = 0, the ball has angular velocity of 10 rads^{-1} and angular position of 4 rad. The most appropriate expression for the angular position of the ball is :

Open solution

(A) $\frac{3}{2}t^4 - t^2 + 10t$

(B) $\frac{t^4}{2} - \frac{t^3}{3} + 10t + 4$

(C) $\frac{2t^4}{3} - \frac{t^3}{6} + 10t + 12$

(D) $2t^4 - \frac{t^3}{2} + 5t + 4$

Q52. MCQ (Single Correct Answer)

A block of mass 2 kg moving on a horizontal surface with speed of 4 ms^{-1} enters a rough surface ranging from x = 0.5 m to x = 1.5 m. The retarding force in this range of rough surface is related to distance by F = $-$kx where k = 12 Nm^{-1}. The speed of the block as it just crosses the rough surface will be :

Open solution

(A) zero

(B) 1.5 ms^{-1}

(C) 2.0 ms^{-1}

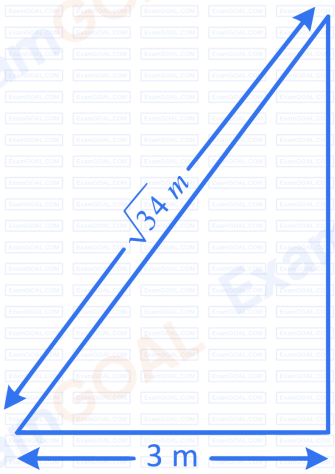
(D) 2.5 ms^{-1}

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Q53. MCQ (Single Correct Answer)

A $\sqrt{34}$ m long ladder weighing 10 kg leans on a frictionless wall. Its feet rest on the floor 3 m away from the wall as shown in the figure. If F_f and F_w are the reaction forces of the floor and the wall, then ratio of F_w/F_f will be :

(Use g = 10 m/s^2.)



Open solution

(A) $\frac{6}{\sqrt{110}}$ (B) $\frac{3}{\sqrt{113}}$ (C) $\frac{3}{\sqrt{109}}$ (D) $\frac{2}{\sqrt{109}}$

Q54. MCQ (Single Correct Answer)

Water falls from a 40 m high dam at the rate of 9×10^4 kg per hour. Fifty percentage of gravitational potential energy can be converted into electrical energy. Using this hydroelectric energy number of 100 W lamps, that can be lit, is :

(Take g = 10 ms^{-2})

Open solution

(A) 25 (B) 50 (C) 100 (D) 18

Q55. MCQ (Single Correct Answer)

Two objects of equal masses placed at certain distance from each other attracts each other with a force of F. If one-third mass of one object is transferred to the other object, then the new force will be :

Open solution

(A) $\frac{2}{9}$ F (B) $\frac{16}{9}$ F (C) $\frac{8}{9}$ F (D) F

Q56. MCQ (Single Correct Answer)

A water drop of radius 1 μm falls in a situation where the effect of buoyant force is negligible. Co-efficient of viscosity of air is 1.8×10^{-5} Nsm^{-2} and its density is negligible as compared to that of water 10^6 gm^{-3}. Terminal velocity of the water drop is :

(Take acceleration due to gravity = 10 ms^{-2})

Open solution

(A) 145.4×10^{-6} ms^{-1} (B) 118.0×10^{-6} ms^{-1}

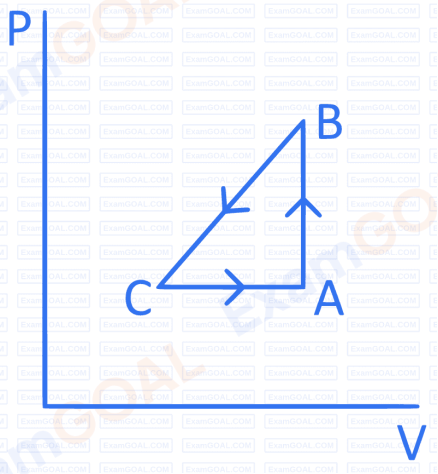
(C) 132.6×10^{-6} ms^{-1} (D) 123.4×10^{-6} ms^{-1}

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Q57. MCQ (Single Correct Answer)

Open solution

A sample of an ideal gas is taken through the cyclic process ABCA as shown in figure. It absorbs, 40 J of heat during the part AB, no heat during BC and rejects 60 J of heat during CA. A work of 50 J is done on the gas during the part BC. The internal energy of the gas at A is 1560 J. The workdone by the gas during the part CA is :



(A) 20 J (B) 30 J (C) −30 J (D) −60 J

Q58. MCQ (Single Correct Answer)

Open solution

What will be the effect on the root mean square velocity of oxygen molecules if the temperature is doubled and oxygen molecule dissociates into atomic oxygen?

(A) The velocity of atomic oxygen remains same

(B) The velocity of atomic oxygen doubles

(C) The velocity of atomic oxygen becomes half

(D) The velocity of atomic oxygen becomes four times

Q59. MCQ (Single Correct Answer)

Open solution

Two point charges A and B of magnitude $+8 \times 10^{-6}$ C and -8×10^{-6} C respectively are placed at a distance d apart. The electric field at the middle point O between the charges is 6.4×10^{4} NC^{-1}. The distance 'd' between the point charges A and B is :

(A) 2.0 m (B) 3.0 m (C) 1.0 m (D) 4.0 m

Q60. MCQ (Single Correct Answer)

Open solution

Resistance of the wire is measured as 2 Ω and 3 Ω at 10°C and 30°C respectively. Temperature co-efficient of resistance of the material of the wire is :

(A) $0.033\ ^{\circ}C^{-1}$ (B) $-0.033\ ^{\circ}C^{-1}$ (C) $0.011\ ^{\circ}C^{-1}$ (D) $0.055\ ^{\circ}C^{-1}$

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Q61. MCQ (Single Correct Answer)

The space inside a straight current carrying solenoid is filled with a magnetic material having magnetic susceptibility equal to 1.2×10^{-5}. What is fractional increase in the magnetic field inside solenoid with respect to air as medium inside the solenoid?

Open solution

(A) 1.2×10^{-5} (B) 1.2×10^{-3} (C) 1.8×10^{-3} (D) 2.4×10^{-5}

Q62. MCQ (Single Correct Answer)

Two parallel, long wires are kept 0.20 m apart in vacuum, each carrying current of x A in the same direction. If the force of attraction per meter of each wire is 2×10^{-6} N, then the value of x is approximately :

Open solution

(A) 1 (B) 2.4 (C) 1.4 (D) 2

Q63. MCQ (Single Correct Answer)

A coil is placed in a time varying magnetic field. If the number of turns in the coil were to be halved and the radius of wire doubled, the electrical power dissipated due to the current induced in the coil would be :

(Assume the coil to be short circuited.)

Open solution

(A) Halved (B) Quadrupled (C) The same (D) Doubled

Q64. MCQ (Single Correct Answer)

An EM wave propagating in x-direction has a wavelength of 8 mm. The electric field vibrating y-direction has maximum magnitude of 60 Vm^{-1}. Choose the correct equations for electric and magnetic fields if the EM wave is propagating in vacuum :

Open solution

(A) $E_y = 60 \sin\left[\frac{\pi}{4} \times 10^3(x - 3 \times 10^8 t)\right]\hat{j}\ Vm^{-1}$

$B_z = 2 \sin\left[\frac{\pi}{4} \times 10^3(x - 3 \times 10^8 t)\right]\hat{k}\ T$

(B) $E_y = 60 \sin\left[\frac{\pi}{4} \times 10^3(x - 3 \times 10^8 t)\right]\hat{j}\ Vm^{-1}$

$B_z = 2 \times 10^{-7} \sin\left[\frac{\pi}{4} \times 10^3(x - 3 \times 10^8 t)\right]\hat{k}\ T$

(C) $E_y = 2 \times 10^{-7} \sin\left[\frac{\pi}{4} \times 10^3(x - 3 \times 10^8 t)\right]\hat{j}\ Vm^{-1}$

$B_z = 60 \sin\left[\frac{\pi}{4} \times 10^3(x - 3 \times 10^8 t)\right]\hat{k}\ T$

(D) $E_y = 2 \times 10^{-7} \sin\left[\frac{\pi}{4} \times 10^4(x - 4 \times 10^8 t)\right]\hat{j}\ Vm^{-1}$

$B_z = 60 \sin\left[\frac{\pi}{4} \times 10^4(x - 4 \times 10^8 t)\right]\hat{k}\ T$

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Q65. MCQ (Single Correct Answer)

Open solution

In Young's double slit experiment performed using a monochromatic light of wavelength λ, when a glass plate (μ = 1.5) of thickness xλ is introduced in the path of the one of the interfering beams, the intensity at the position where the central maximum occurred previously remains unchanged. The value of x will be :

(A) 3 (B) 2 (C) 1.5 (D) 0.5

Q66. MCQ (Single Correct Answer)

Open solution

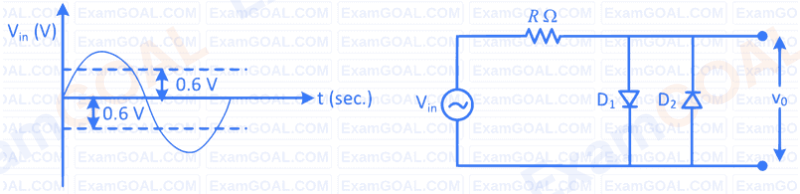
Let K_1 and K_2 be the maximum kinetic energies of photo-electrons emitted when two monochromatic beams of wavelength λ_1 and λ_2, respectively are incident on a metallic surface. If $\lambda_1 = 3\lambda_2$ then :

(A) $K_1 > \frac{K_2}{3}$ (B) $K_1 < \frac{K_2}{3}$ (C) $K_1 = \frac{K_2}{3}$ (D) $K_2 = \frac{K_1}{3}$

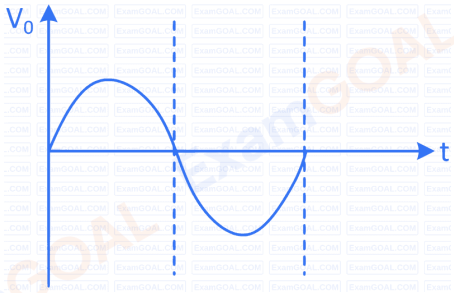
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Q67. MCQ (Single Correct Answer)

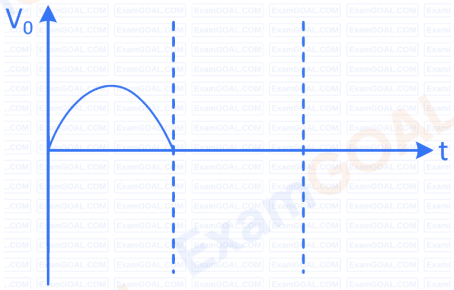
In the given circuit the input voltage V_{in} is shown in figure. The cut-in voltage of p-n junction diode (D_1 or D_2) is 0.6 V. Which of the following output voltage (V_0) waveform across the diode is correct?



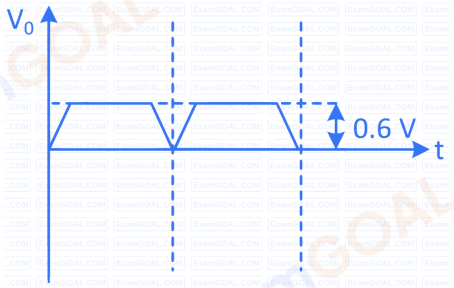
(A)



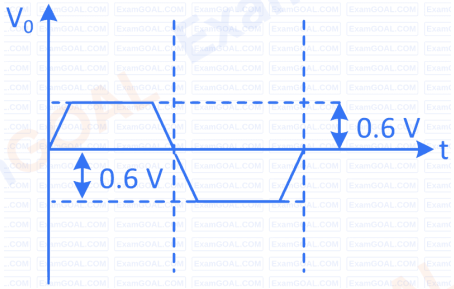
(B)



(C)



(D)



Open solution

Q68. Numerical

A student in the laboratory measures thickness of a wire using screw gauge. The readings are 1.22 mm, 1.23 mm, 1.19 mm and 1.20 mm. The percentage error is $\frac{x}{121}\%$. The value of x is _____.

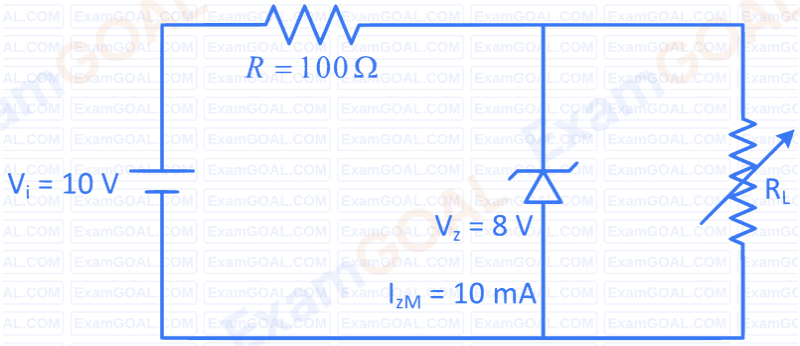
Open solution

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Q69. Numerical

A zener of breakdown voltage V_z = 8 V and maximum zener current, I_{ZM} = 10 mA is subjected to an input voltage V_i = 10 V with series resistance R = 100 Ω. In the given circuit R_L represents the variable load resistance. The ratio of maximum and minimum value of R_L is _____.

Open solution



Q70. Numerical

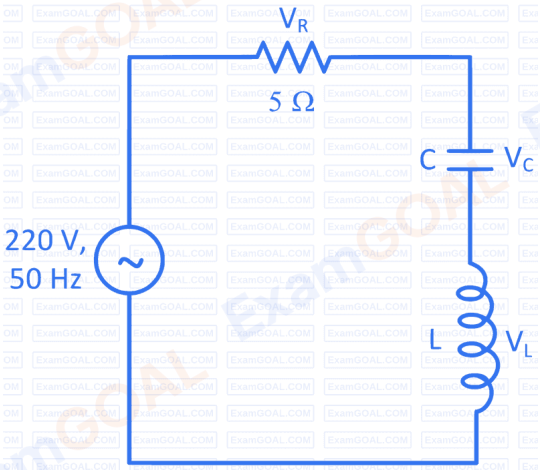
In a Young's double slit experiment, an angular width of the fringe is 0.35° on a screen placed at 2 m away for particular wavelength of 450 nm. The angular width of the fringe, when whole system is immersed in a medium of refractive index 7/5, is $\frac{1}{\alpha}$. The value of α is _____.

Open solution

Q71. Numerical

In the given circuit, the magnitude of V_L and V_C are twice that of V_R. Given that f = 50 Hz, the inductance of the coil is $\frac{1}{K\pi}$ mH. The value of K is _____.

Open solution

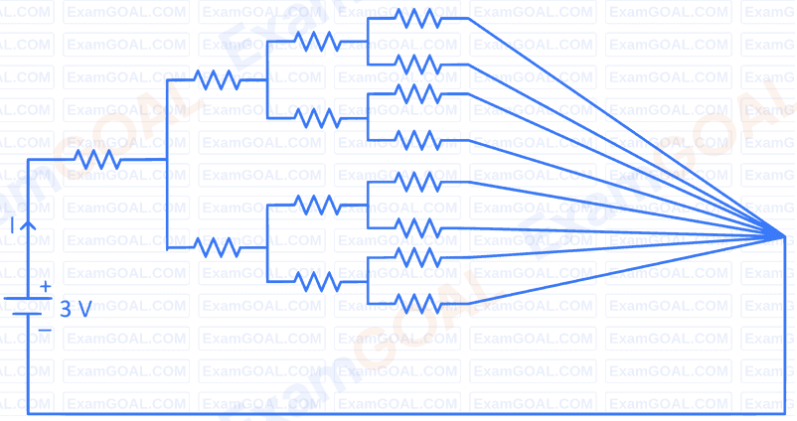


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Q72. Numerical

All resistances in figure are 1 Ω each. The value of current 'I' is $\frac{a}{5}$ A. The value of a is _____.

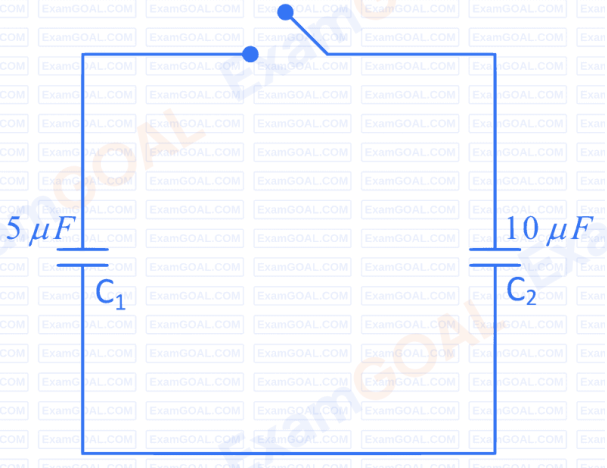
Open solution



Q73. Numerical

A capacitor C_1 of capacitance 5 μF is charged to a potential of 30 V using a battery. The battery is then removed and the charged capacitor is connected to an uncharged capacitor C_2 of capacitance 10 μF as shown in figure. When the switch is closed charge flows between the capacitors. At equilibrium, the charge on the capacitor C_2 is _____ μC.

Open solution



Q74. Numerical

A tuning fork of frequency 340 Hz resonates in the fundamental mode with an air column of length 125 cm in a cylindrical tube closed at one end. When water is slowly poured in it, the minimum height of water required for observing resonance once again is _____ cm.

(Velocity of sound in air is 340 ms^{-1})

Open solution

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Q75. Numerical

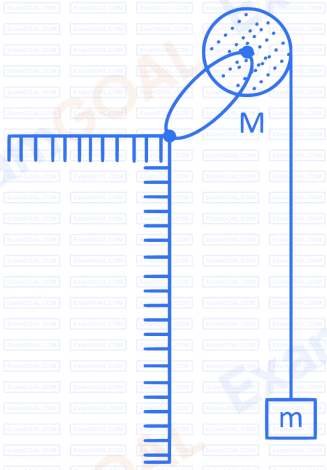
A liquid of density 750 kgm^{-3} flows smoothly through a horizontal pipe that tapers in cross-sectional area from $A_1 = 1.2 \times 10^{-2}\ m^2$ to $A_2 = \frac{A_1}{2}$. The pressure difference between the wide and narrow sections of the pipe is 4500 Pa. The rate of flow of liquid is _____ $\times 10^{-3}\ m^3s^{-1}$.

Open solution

Q76. Numerical

A uniform disc with mass M = 4 kg and radius R = 10 cm is mounted on a fixed horizontal axle as shown in figure. A block with mass m = 2 kg hangs from a massless cord that is wrapped around the rim of the disc. During the fall of the block, the cord does not slip and there is no friction at the axle. The tension in the cord is _____ N.

(Take g = 10 ms^{-2})



Open solution

Q77. Numerical

A car covers AB distance with first one-third at velocity v_1 ms^{-1}, second one-third at v_2 ms^{-1} and last one-third at v_3 ms^{-1}. If $v_3 = 3v_1$, $v_2 = 2v_1$ and $v_1 = 11$ ms^{-1} then the average velocity of the car is _____ ms^{-1}.



Open solution

Out of syllabus 13 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

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Q78. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Match List - I with List - II :

	List-I		List-II
(A)	negatively charged sol	(I)	$Fe_2O_3 \, . \, xH_2O$
(B)	macromolecular colloid	(II)	CdS sol
(C)	positively charged sol	(III)	Starch
(D)	Cheese	(IV)	a gel

Choose the correct answer from the options given below:

(A) (A) - (II), (B) - (III), (C) - (IV), (D) - (I)

(B) (A) - (II), (B) - (I), (C) - (III), (D) - (IV)

(C) (A) - (II), (B) - (III), (C) - (I), (D) - (IV)

(D) (A) - (I), (B) - (III), (C) - (II), (D) - (IV)

Q79. MCQ (Single Correct Answer) · **Chemistry**

Open solution

In the metallurgical extraction of copper, following reaction is used :

$FeO + SiO_2 \rightarrow FeSiO_3$

FeO and $FeSiO_3$ respectively are.

(A) gangue and flux. **(B)** flux and slag. **(C)** slag and flux. **(D)** gangue and slag.

Q80. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Hydrogen has three isotopes : protium (^1H), deuterium (^2H or D) and tritium (^3H or T). They have nearly same chemical properties but different physical properties. They differ in :

(A) number of protons. **(B)** atomic number.

(C) electronic configuration. **(D)** atomic mass.

Q81. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Among the given oxides of nitrogen ; N_2O, N_2O_3, N_2O_4 and N_2O_5, the number of compound/(s) having N - N bond is :

(A) 1 **(B)** 2 **(C)** 3 **(D)** 4

Q82. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Which of the following oxoacids of sulphur contains "S" in two different oxidation states?

(A) $H_2S_2O_3$ **(B)** $H_2S_2O_6$ **(C)** $H_2S_2O_7$ **(D)** $H_2S_2O_8$

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Q83. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Correct statement about photo-chemical smog is :

(A) It occurs in humid climate.

(B) It is a mixture of smoke, fog and SO_2.

(C) It is reducing smog.

(D) It results from reaction of unsaturated hydrocarbons.

Q84. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Given below are two statements, one is Assertion (A) and other is Reason (R).

Assertion (A) : Natural rubber is a linear polymer of isoprene called cis-polyisoprene with elastic properties.

Reason (R) : The cis-polyisoprene molecules consist of various chains held together by strong polar interactions with coiled structure.

In the light of the above statements, choose the correct one from the options given below :

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

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Q85. MCQ (Single Correct Answer) · **Chemistry**

Open solution

The drug tegamet is :

(A)



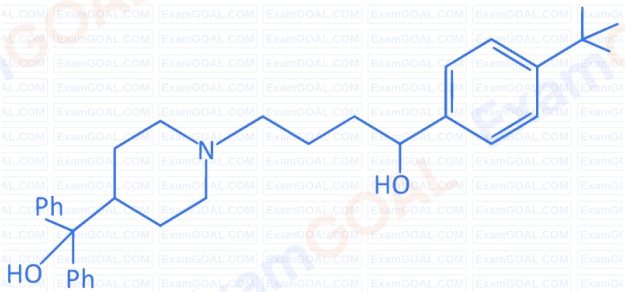
(B)



(C)



(D)



Q86. Numerical · **Chemistry**

Open solution

100 g of an ideal gas is kept in a cylinder of 416 L volume at 27°C under 1.5 bar pressure. The molar mass of the gas is _____ g mol^{-1}. (Nearest integer)

(Given : R = 0.083 L bar K^{-1} mol^{-1})

Q87. MCQ (Single Correct Answer) · **Mathematics**

Open solution

Let the plane ax + by + cz = d pass through (2, 3, −5) and is perpendicular to the planes 2x + y − 5z = 10 and 3x + 5y − 7z = 12. If a, b, c, d are integers d > 0 and gcd (|a|, |b|, |c|, d) = 1, then the value of a + 7b + c + 20d is equal to :

(A) 18 (B) 20 (C) 24 (D) 22

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Q88. Numerical · **Mathematics**

The maximum number of compound propositions, out of $p \vee r \vee s$, $p \vee r \vee \sim s$, $p \vee \sim q \vee s$, $\sim p \vee \sim r \vee s$, $\sim p \vee \sim r \vee \sim s$, $\sim p \vee q \vee \sim s$, $q \vee r \vee \sim s$, $q \vee \sim r \vee \sim s$, $\sim p \vee \sim q \vee \sim s$ that can be made simultaneously true by an assignment of the truth values to p, q, r and s, is equal to _____.

Open solution

Q89. MCQ (Single Correct Answer) · **Physics**

Following statements related to radioactivity are given below :

(A) Radioactivity is a random and spontaneous process and is dependent on physical and chemical conditions.

(B) The number of un-decayed nuclei in the radioactive sample decays exponentially with time.

(C) Slope of the graph of $\log_e$ (no. of undecayed nuclei) Vs. time represents the reciprocal of mean life time (τ).

(D) Product of decay constant (λ) and half-life time ($T_{1/2}$) is not constant.

Choose the most appropriate answer from the options given below :

(A) (A) and (B) only **(B)** (B) and (D) only **(C)** (B) and (C) only **(D)** (C) and (D) only

Open solution

Q90. MCQ (Single Correct Answer) · **Physics**

Amplitude modulated wave is represented by $V_{AM} = 10\left[1 + 0.4\cos(2\pi \times 10^4 t)\right]\cos(2\pi \times 10^7 t)$. The total bandwidth of the amplitude modulated wave is :

(A) 10 kHz **(B)** 20 MHz **(C)** 20 kHz **(D)** 10 MHz

Open solution

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Answer Key

Chemistry

1. **D**	2. **C**	3. **A**	4. **B**	5. **B**
6. **C**	7. **C**	8. **C**	9. **C**	10. **C**
11. **C**	12. **B**	13. **600**	14. **330**	15. **16**
16. **75**	17. **2**	18. **1**	19. **46**	20. **1**
21. **30**				

Mathematics

22. **B**	23. **A**	24. **D**	25. **B**	26. **C**
27. **B**	28. **B**	29. **C**	30. **C**	31. **D**
32. **B**	33. **B**	34. **D**	35. **D**	36. **B**
37. **D**	38. **C**	39. **C**	40. **A**	41. **125**
42. **0**	43. **10**	44. **11**	45. **41651**	46. **58**
47. **25**	48. **2**	49. **37**		

Physics

50. **A**	51. **B**	52. **C**	53. **C**	54. **B**
55. **C**	56. **D**	57. **B**	58. **B**	59. **B**
60. **A**	61. **A**	62. **C**	63. **D**	64. **B**
65. **B**	66. **B**	67. **D**	68. **150**	69. **2**
70. **4**	71. **0**	72. **8**	73. **100**	74. **50**
75. **24**	76. **10**	77. **18**		

Out of syllabus

78. **C**	79. **D**	80. **D**	81. **C**	82. **A**
83. **D**	84. **C**	85. **C**	86. **4**	87. **D**
88. **9**	89. **C**	90. **C**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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